

MUKESH M

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PROFESSIONAL SUMMARY

Enthusiastic and detail-oriented AI & ML fresher with hands-on experience in Machine Learning, Deep Learning, and Explainable AI. Skilled in Python, TensorFlow, PyTorch, Scikit-learn, and CNN-based model development for healthcare and computer vision applications. Strong understanding of data preprocessing, feature engineering, model optimization, and performance evaluation techniques. Experienced in developing end-to-end AI solutions with a focus on accuracy, scalability, and real-world problem solving. Quick learner with strong analytical, research, and problem-solving abilities, passionate about building innovative and impactful AI systems.

TECHNICAL SKILLS

Programming : Python , SQL
Frameworks: TensorFlow, PyTorch, Scikit-learn
Deep Learning : CNN ,Transfer Learning , Data Augmentation, Model Evaluation.

Deployment & Tools : Power BI, Tableau ,MySQL Workbench, ONNX, Git, GitHub
Concepts : Machine Learning, Deep Learning, Explainable AI (XAI), Feature Engineering, Model inference Optimization

PROJECTS

Disease Classification and Recommendation System using CNN and Geo-Demographic Data : (Published)

- Developed an end-to-end deep learning system ranging from preprocessing to feature extraction and model training. Compared the performance of CNN with SVM, Random Forest, and Decision Tree classifiers.
- Reached a classification accuracy of ~95% with better generalization using validation methods. Designed a real-time prediction model combining geo-demographic information with forecasting models for healthcare analysis.

Lung Disease Detection and Recommendation System Using Explainable AI with Metadata : (submitted for publication)

- Developed an X-ray image classification model using CNN and metadata fusion, optimized with ONNX for fast CPU-based inference.
- Achieved 96% accuracy with F1-score of 0.93 on 5,000+ labelled chest X-ray images using stratified train-test split. Applied Explainable AI (Grad-CAM) to generate visual interpretability maps, improving model transparency and decision traceability.
- Integrated a rule-based recommendation engine to deliver symptom analysis, treatment suggestions, and clinical decision support outputs. data set used NIH ChestX-ray14, Kaggle dataset .

EDUCATION

Vellore Institute of Technology (VIT)

- MTech - CSE(ArtificialIntelligenceandMachine Learning)

Pursuing 2024-26
(CGPA : 7.99/10)

KINGSTON Engineering College

- **BE-Mechanical Engineering**
(RelevantCoursework:AutoCAD,Nx-cadd, SolidWorks, Catia and Ansys)

May 2023
(CGPA : 8.9/10)

ADDITIONAL INFORMATION

1. **Languages** : English and Tamil

2. **Certifications :**

- **Masterin data Science&AnalyticswithAI:** Certified from I.T.Vedant, demonstrating core data analytics knowledge.
- **Fundamentals MachinelearningwithAI:** Hands-on experience in building and evaluating ML models regression, classification, (clustering). Skilled in data pre-processing using libraries like Scikit-learn, TensorFlow, Pandas and NumPy.